

Education

PhD, Australian National University, Australia

Expected 2028

- **Research Interests:** 3D Vision, Computer Graphics, 3D Generative Models, Geometric Deep Learning, and Articulated Objects.
- **Supervisors:** Prof. Hongdong Li, Prof. Richard Hartley, and A/Prof. Mehrtaash Harandi.

Master of Machine Learning and Computer Vision, Australian National University, Australia

2023

- Thesis: *Generative Model Approaches for 3D Full-Body Human Pose Reconstruction from Sparse Joint Observations.*

Bachelor of Electrical Engineering, University of Sydney, Australia

2020

- Thesis: *Wireless Signal Modulation Classification with Deep Learning Methods.*

Selected Research Projects

End-to-End Articulated 3D Object Reconstruction

- We study articulated 3D object reconstruction from sparse observations of two articulation states. We leverage a learned geometry prior from static 3D reconstruction to recover complete object geometry, while jointly predicting movable-part segmentation and articulation parameters in a unified feed-forward framework. Our method performs reconstruction in **0.32 seconds per object**, compared with approximately **9 minutes** required by prior per-object optimization-based approaches.
- **Publication:** Xuyang Wang, Zhenyu Li, Jian Ding, Habib Slim, Peter Wonka, Hongdong Li, Mohammed Elhoseiny. “**Artic-O: End-to-End Articulated Object Reconstruction via Latent Geometry Learning.**” Conditionally accepted, ACM SIGGRAPH Asia 2026 Conference Papers.

Generative Learning on 3D Meshes with Manifold Diffusion

- We study generative modeling of signals directly on 3D mesh surfaces by representing meshes as discrete manifolds and surface attributes as functions defined on them. We propose **DoubleDiffusion**, which combines geometry-aware heat diffusion through the Laplace–Beltrami operator with denoising diffusion, enabling UV-free generative learning directly on irregular meshes and scaling to meshes with up to **100K vertices**.
- **Publication:** Xuyang Wang, Ziang Cheng, Zhenyu Li, Jiayu Yang, Haorui Ji, Pan Ji, Mehrtaash Harandi, Richard Hartley, Hongdong Li. “**DoubleDiffusion: Combining Heat Diffusion with Denoising Diffusion for Generative Learning on 3D Meshes.**” Preprint; [Project Page](#).

Monocular Depth Estimation with Adaptive Bins

- We study monocular depth estimation with a DETR-style decoder that progressively predicts adaptive depth bins and estimates depth maps in a coarse-to-fine manner.
- **Publication:** Zhenyu Li, Xuyang Wang, Junjun Jiang, Xianming Liu. “**BinsFormer: Revisiting Adaptive Bins for Monocular Depth Estimation.**” *IEEE Transactions on Image Processing*, 2024. [Project Page](#).

Research and Industry Experience

Visiting PhD Student

Vision-CAIR, KAUST, Saudi Arabia

Jan. 2026 – Dec. 2026

- Conducting research on articulated object reconstruction, 3D vision, and generative models, with a focus on efficient reconstruction of articulated 3D assets from sparse image observations.
- Developed the Artic-O feed-forward framework for jointly recovering complete object geometry, movable-part segmentation, and articulation parameters using learned 3D geometry priors.

Research Intern

Tencent XR Vision Lab, Canberra, Australia

Aug. 2022 – May. 2023

- Developed generative models for reconstructing full-body SMPL poses from sparse signals captured by a VR headset and hand controllers, exploring Transformer-, VAE/CVAE, and diffusion-based architectures.

Software Engineer Intern

Baidu Inc., Shanghai, China

Nov. 2021 – Jan. 2022

- Developed a 3D point-cloud tracking pipeline for autonomous driving using nuScenes and internal datasets, including point-cloud visualization, data preprocessing, and neural object tracking based on CenterPoint.

Research Intern

Corerain Technologies, Shenzhen, China

Dec. 2019 – Feb. 2020

- Investigated hardware-level parallelism and acceleration opportunities for RNN, GRU, LSTM, and attention-based architectures.

Academic Service

Academic Tutor

Australian National University

2022 – 2024

- COMP8610 Computer Graphics, Semester 1, 2024. Supervisor: Prof. Hongdong Li.
- COMP8536 Advanced Topics in Deep Learning in Computer Vision, Semester 2, 2022. Supervisor: Prof. Nick Barnes.

Conference Reviewer: ECCV 2024, ECCV Workshops 2024, NeurIPS 2024, CVPR 2025, ICML 2025.

Awards and Scholarships

University Research Scholarship, Australian National University

Jan. 2024 – Jan. 2028

- Scholarship for international higher-degree research candidates, associated with the ANU HDR Fee Merit Scholarship covering full PhD tuition fees.

HEX Singapore Program Scholarship

Jun. 2023 – Jul. 2023

- Awarded by the Australian National University to support participation in the HEX Singapore program, including return airfare, accommodation, and living allowance.

Vice-Chancellor’s Global Mobility Scholarship, University of Sydney

Jul. 2019 – Aug. 2019

- Supported participation in the summer program *Global Cities as Centers of Knowledge Production* at Humboldt University of Berlin.